Extra Dessert #8: Satellite Operations

Transponders (linear and inverting), satellite modes, Keplerian elements, store-and-forward, orbit types, and antenna polarization.

Group E2A · 12 pool questions · Self-study reference · Practice exams at eham.net/exams

Orbit Vocabulary

Question	Answer	Q
Direction of an ascending pass	**From south to north**	E2A01
Keplerian elements	**Parameters that define the orbit of a satellite**	E2A06
Satellite that appears to stay in one position	**Geostationary**	E2A10

Transponders

Question	Answer	Q
Inverting linear transponder characteristics	Doppler shift is reduced (uplink and downlink shifts are in opposite directions) + signal position in the band is reversed + USB on uplink becomes LSB on downlink – **all of the above**	E2A02
How an inverting transponder processes uplink	**Signal is mixed with a local oscillator signal and the difference product is transmitted**	E2A03
Signals supported by a linear transponder	FM, CW, SSB, SSTV, PSK, packet – **all of the above**	E2A07
Why limit ERP into a linear-transponder satellite	**To avoid reducing the downlink power to all other users** (one strong signal drives the linear amplifier into compression, stealing output power from all others)	E2A08
Purpose of store-and-forward	**To hold digital messages in the satellite for later download** (when over the destination)	E2A12

Satellite Modes And Bands

Question	Answer	Q
What is the “mode” of a satellite?	**The satellite’s uplink and downlink frequency bands**	E2A04
What letters in a mode designator specify	**The uplink and downlink frequency ranges**	E2A05
“L band” and “S band”	**The 23-centimeter and 13-centimeter bands** (respectively)	E2A09

Antennas For Satellite Work

Question	Answer	Q
Best antenna to minimize spin modulation and Faraday rotation	**A circularly polarized antenna**	E2A11

Dessert 8 Question Bank (E2A)

E2A01. What is the direction of an ascending pass for an amateur satellite?
A. From west to east.
B. From east to west.
C. From south to north.
D. From north to south.

E2A02. Which of the following is characteristic of an inverting linear transponder?
A. Doppler shift is reduced because the uplink and downlink shifts are in opposite directions.
B. Signal position in the band is reversed.
C. Upper sideband on the uplink becomes lower sideband on the downlink, and vice versa.
D. All these choices are correct.

E2A03. How is an upload signal processed by an inverting linear transponder?
A. The signal is detected and remodulated on the reverse sideband.
B. The signal is passed through a nonlinear filter.
C. The signal is reduced to I and Q components, and the Q component is filtered out.
D. The signal is mixed with a local oscillator signal and the difference product is transmitted.

E2A04. What is meant by the “mode” of an amateur radio satellite?
A. Whether the satellite is in a low earth or geostationary orbit.
B. The satellite’s uplink and downlink frequency bands.
C. The satellite’s orientation with respect to the Earth.
D. Whether the satellite is in a polar or equatorial orbit.

E2A05. What do the letters in a satellite’s mode designator specify?
A. Power limits for uplink and downlink transmissions.
B. The location of the ground control station.
C. The polarization of uplink and downlink signals.
D. The uplink and downlink frequency ranges.

E2A06. What are Keplerian elements?
A. Parameters that define the orbit of a satellite.
B. Phase reversing elements in a Yagi antenna.
C. High-emission heater filaments used in magnetron tubes.
D. Encrypting codes used for spread spectrum modulation.

E2A07. Which of the following types of signals can be relayed through a linear transponder?
A. FM and CW.
B. SSB and SSTV.
C. PSK and packet.
D. All these choices are correct.

E2A08. Why should effective radiated power (ERP) be limited to a satellite that uses a linear transponder?
A. To prevent creating errors in the satellite telemetry.
B. To avoid reducing the downlink power to all other users.
C. To prevent the satellite from emitting out-of-band signals.
D. To avoid interfering with terrestrial QSOs.

E2A09. What do the terms “L band” and “S band” specify?
A. The 23- and 13-centimeter bands.
B. The 2-meter and 70-centimeter bands.
C. FM and digital store-and-forward systems.
D. Which sideband to use.

E2A10. What type of satellite appears to stay in one position in the sky?
A. HEO.
B. Geostationary.
C. Geomagnetic.
D. LEO.

E2A11. What type of antenna can be used to minimize the effects of spin modulation and Faraday rotation?
A. A linearly polarized antenna.
B. A circularly polarized antenna.
C. An isotropic antenna.
D. A log-periodic dipole array.

E2A12. What is the purpose of digital store-and-forward functions on an amateur radio satellite?
A. To upload operational software for the transponder.
B. To delay download of telemetry between satellites.
C. To hold digital messages in the satellite for later download.
D. To relay messages between satellites.

Answer Key

E2A01: C	E2A02: D	E2A03: D	E2A04: B
E2A05: D	E2A06: A	E2A07: D	E2A08: B
E2A09: A	E2A10: B	E2A11: B	E2A12: C